PROPOSED DIGITAL PUBLIC INFRASTRUCTURE (DPI)

Green Credit Programme: Automation Engine

A Long-Term Verification & Allocation Platform for MoEFCC's 2023 Initiative

Event: CU Project Expo 2026

TEAM MEMBERS:

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PROBLEM STATEMENT

The Green Credit Programme (initiated late 2023) is currently in its infancy. Scaling it nationwide faces critical bottlenecks:

- ✗ **Manual Allocation:** Fragmented state-level data makes finding degraded land difficult.
- ✗ **Opaque Verification:** Reliance solely on physical officer visits is slow and prone to corruption.
- ✗ **Delayed Incentives:** Credit issuance can take 12-24 months, discouraging corporate CSR participation.

OBJECTIVES

To build the foundational digital infrastructure required to scale the GCP over the next decade by:

- ✓ Centralizing the national inventory of degraded lands using GIS.
- ✓ Automating verification using **AI & Satellite Imagery** .
- ✓ Creating a transparent, immutable ledger for Green Credit issuance and exchange.

IMPACT INFOGRAPHICS

PROCESS EFFICIENCY COMPARISON

TRADITIONAL FLOW	AUTOMATED FLOW
2+ Yrs	< 90 Days
Manual officer audits	AI-driven verification

SYSTEM TRUST SCORE

99.9%

Auditability & Transparency

Achieved via automated Sentinel-2 satellite ingestion and immutable ledgering, removing human bias.

METHODOLOGY & BLOCK DIAGRAM

The system employs a microservices architecture, integrating corporate user interfaces with government spatial data and automated AI verification pipelines.

1. PRESENTATION LAYER (REACT SPA)

- **Company Portal** — Auth & Profiles
- **Land Allocation** — GIS Map Interfaces
- **Credits Dashboard** — Ledger & Redemption

2. CORE LOGIC LAYER (NODE.JS/PYTHON)

- **API Gateway** — REST Routing & RBAC
- **Data Aggregator** — Multi-Ministry Mapping

3. PERSISTENCE & AI ENGINE

DATA PERSISTENCE

- **PostGIS Database** — Spatial Land Grid Arrays
- **PostgreSQL Ledger** — Immutable Transactions

AI VERIFICATION CORE

- Ingest Sentinel-2 Pipeline
- Compute NDVI & Biomass
- Map Canopy Density (%)
- Output Validation Score

DATA INGESTION ENDPOINTS: Sentinel APIs | MoEFCC Hub | IoT Telemetry

CREDIT CALCULATION

The programmatic logic converting physical restoration into digital credits.

```
/* 1. Ecological Perf. Score */
EPS = (Canopy% × W₁) + (SoilC × W₂)
/* 2. Final Credit Value */
Total_GC = Area × CpH × EPS × VCM
```

*CpH: Credits-per-Hectare coefficient
*VCM: Verification Confidence Multiplier (AI)

OUTCOMES & FUTURE SCOPE

IMMEDIATE OUTCOMES

- ✓ **Accelerated CSR:** Connects corporate capital directly to degraded land.
- ✓ **Automated Trust:** Removes human subjectivity from verification.

LONG-TERM VISION (2026-2030)

Because the GCP is newly initiated, this architecture is built to scale drastically over the next decade:

- **Blockchain Integration** — Moving the PostgreSQL ledger to a permissioned blockchain (Hyperledger).
- **Carbon Market Linking** — Enabling Green Credits to be traded on international offset marketplaces.
- **High-Res Drone LiDAR** — Upgrading from Sentinel-2 satellite data to 3D LiDAR for precise biomass.